



100 Days of Python for Data Analytics

Welcome to my **100-Day Data Analytics Journey**.

This repository documents my commitment to becoming a skilled **Data Analyst** using **Python, SQL, Statistics, Data Visualization, and Real-World Analytics Projects**.

For the next 100 days, I will learn, practice, build, and share my progress publicly through daily code, notes, exercises, and projects.



Goal

By the end of this challenge, I will be able to:

- Write clean and efficient Python code
 - Manipulate and analyze data using Pandas and NumPy
 - Create insightful visualizations
 - Perform Exploratory Data Analysis (EDA)
 - Apply statistical concepts to business problems
 - Write advanced SQL queries
 - Build dashboards and reports
 - Complete end-to-end data analytics projects
 - Develop a professional analytics portfolio
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Tech Stack

Programming

- Python

Data Analysis

- NumPy
- Pandas

Data Visualization

- Matplotlib
- Seaborn
- Plotly

Database

- SQL
- PostgreSQL

Dashboarding

- Power BI
- Tableau

Development Tools

- Jupyter Notebook
- VS Code
- Git
- GitHub

 Roadmap

Phase	Days	Topics
Python Fundamentals	1-20	Python Basics, OOP, Functions, File Handling
NumPy	21-30	Arrays, Operations, Statistics
Pandas	31-50	Data Cleaning, Transformation, EDA
Visualization	51-65	Matplotlib, Seaborn, Plotly
Statistics	66-75	Probability, Hypothesis Testing, A/B Testing
SQL	76-85	Queries, Joins, Window Functions
Projects	86-100	Real-World Data Analytics Projects

 Repository Structure

```
100-days-data-analytics-python/
|
|— README.md
|— resources/
|— datasets/
|— notes/
|
|— day01/
|— day02/
|— day03/
|
|...
|— day100/
|
|— projects/
|   |— project01_sales_analysis/
|   |— project02_customer_segmentation/
|   |— project03_marketing_analysis/
```

```
|  
|├─ project04_hr_analytics/  
|└─ capstone_project/  
└─ portfolio/
```



Daily Folder Structure

Each day follows a consistent structure:

```
dayXX/  
|  
|├─ README.md  
|├─ notes.md  
|├─ exercises.py  
|├─ practice.ipynb  
|├─ solutions.py  
|└─ datasets/
```



Projects

Throughout the challenge, I will complete multiple portfolio-worthy projects:

Project 1: Sales Analysis

- Business KPI Analysis
- Revenue Trends
- Product Performance

Project 2: Customer Segmentation

- Customer Behavior Analysis
- Segmentation Techniques
- Insights & Recommendations

Project 3: Marketing Campaign Analysis

- Campaign Performance
- ROI Measurement
- Conversion Funnel Analysis

Project 4: HR Analytics

- Employee Attrition Analysis
- Workforce Metrics

- Predictive Insights

Capstone Project

Choose one:

- Netflix Analytics
- Amazon Sales Analytics
- IPL Analytics
- Blinkit Analytics
- Swiggy Analytics
- Financial Analytics



Skills Covered

Python

- Variables
- Data Types
- Functions
- OOP
- File Handling

Data Analysis

- NumPy
- Pandas
- Data Cleaning
- Feature Engineering

Statistics

- Probability
- Sampling
- Hypothesis Testing
- A/B Testing

SQL

- Joins
- Subqueries
- CTEs
- Window Functions

Visualization

- Matplotlib
- Seaborn

- Plotly

Business Analytics

- KPI Analysis
 - Customer Analytics
 - Marketing Analytics
 - HR Analytics
-



Progress Tracker

Phase 1 – Python Fundamentals

- ☐ Day 01
 - ☐ Day 02
 - ☐ Day 03
 - ☐ Day 04
 - ☐ Day 05
 - ☐ Day 06
 - ☐ Day 07
 - ☐ Day 08
 - ☐ Day 09
 - ☐ Day 10
 - ☐ Day 11
 - ☐ Day 12
 - ☐ Day 13
 - ☐ Day 14
 - ☐ Day 15
 - ☐ Day 16
 - ☐ Day 17
 - ☐ Day 18
 - ☐ Day 19
 - ☐ Day 20
-

Phase 2 – NumPy

- ☐ Day 21–30

Phase 3 – Pandas

- ☐ Day 31–50

Phase 4 – Visualization

- ☐ Day 51–65

Phase 5 – Statistics

- ☐ Day 66–75

Phase 6 – SQL

- ☐ Day 76–85

Phase 7 – Projects

- ☐ Day 86–100



Recommended Resources

Python

- Python Documentation
- Automate the Boring Stuff with Python
- Real Python

Data Analytics

- Kaggle
- DataCamp
- Analytics Vidhya

SQL

- SQLBolt
- Mode Analytics SQL Tutorial

Statistics

- StatQuest
- Khan Academy



Connect With Me

If you're also learning Data Analytics, feel free to:

★ Star this repository

🍴 Fork it

👥 Share your progress

💡 Suggest improvements

Challenge Commitment

Consistency beats intensity.

One day at a time. One notebook at a time. One project at a time.

Let's build a strong Data Analytics portfolio in 100 days. 🚀